Project Workflow: Week 1

Phase: Project Planning & Concept Finalization

1. Objective

The primary goal of Week 1 was to establish a solid foundation for a simple 2D top-down car racing game built entirely in Java using Swing and AWT graphics. This included:

1. Defining the project scope, core gameplay loop, and win/lose conditions.
2. Selecting the right technology for smooth real-time rendering.
3. Planning the road system, environment design, and overall visual layout of the game.

2. Detailed Task Breakdown

Feature Identification

Decided on the core feature set: a 3-lane scrolling road with curve physics, player lane-switching with speed boosting, randomized obstacle cars (EnemyCar), gas pickups, a scoring system, and a 5,000-unit distance win condition.

Technical Stack Selection

Selected Java SE with Swing and AWT graphics for all rendering, requiring no external libraries or game engine. A javax.swing.Timer firing at approximately 60 FPS was chosen to drive the core update/repaint game loop.

Game Layout Planning

Planned the 900x700 game window with a 3-lane road and animated lane markings. The background environment will include trees, bushes, mountains, clouds, lakes, and parks. Also planned a HUD overlay for score, speed, and distance, and a dynamic sky that transitions from dawn to afternoon to dusk as the player progresses.

Class Structure Planning

Planned a modular, object-oriented class structure: a JFrame entry point, a GamePanel for the core loop and rendering pipeline, an Environment class composed of scenery classes (Bird, Cloud, Mountain, Lake, Park, Tree, Bush), a Road class for curve physics, an abstract Car class extended by PlayerCar and EnemyCar, and a GasItem class for collectible pickups.

3. Deliverables

1. Project Scope Document: A written outline of the game's planned features and limitations.
2. Project Timeline: A structured 10-week development roadmap.

4. Tools & Resources

Development Environment

1. Java JDK 8+
2. IntelliJ IDEA / VS Code

Reference Materials

Java Swing, AWT 2D graphics, and javax.swing.Timer-based game loop documentation.